

A Phonological Plausibility Framework for Evaluating Unknown or Ambiguous Symbol Strings

Abstract

This note proposes a neutral, falsification-oriented framework for evaluating how plausibly a given string of symbols could correspond to spoken language systems at the level of language families. Rather than attempting decipherment or semantic interpretation, the framework assesses phonological *compatibility*: the degree to which a symbol string can be accommodated by known phonological constraints with minimal distortion. The method is intended as an auxiliary tool for hypothesis testing in historical linguistics, undeciphered script studies, and comparative phonology.

1. Motivation and scope

A recurring problem in linguistics and epigraphy is the evaluation of unknown, partially known, or ambiguous symbol strings. These include undeciphered scripts, fragmentary inscriptions, coded texts, or artificial symbol systems. In many such cases, traditional methods of decipherment are blocked by the absence of long texts, bilinguals, or clear semantic anchors.

This framework addresses a narrower and more tractable question:

Given a string of symbols, what kinds of phonological systems could plausibly host it?

The framework does **not** attempt translation, decipherment, or identification of a specific language. Its goal is, instead, to provide **negative and comparative evidence**, thereby allowing researchers to reject implausible hypotheses and to rank broader language-family-level phonological compatibility.

2. Conceptual background

Languages differ systematically in:

- Phoneme inventories
- Permissible syllable structures
- Phonotactic constraints
- Preferred repair strategies for foreign material

These differences strongly constrain how foreign or unfamiliar sound sequences are perceived, adapted, or rejected. Loanword adaptation studies demonstrate that phonological systems respond predictably to illegal structures by substituting sounds, inserting vowels, deleting segments, or restructuring syllables.

The proposed framework leverages these well-established properties, extending them to situations where the *source phonology is unknown* and must be hypothesised.

3. Inputs and outputs

3.1 Inputs

- A symbol string (known or unknown orthography)
- A set of phonological profiles representing language families or languages
- Optional metadata (for example, script type, directionality, segmentation hypotheses)

3.2 Outputs

- A list of language families ranked by phonological plausibility
- Qualitative descriptions of required distortions or repairs
- Optional illustrative sound-shapes (non-authoritative)

3.3 Explicit non-goals

- Decipherment
 - Semantic interpretation
 - Identification of specific languages
 - Historical or genealogical claims
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4. Core process overview

The framework consists of five stages, each designed to minimise unjustified assumptions.

Stage 1: Structural Abstraction

The input string is analysed independently of sound values to extract structural properties such as:

- Symbol frequency
- Repetition patterns
- Positional constraints
- Relative symbol counts

The result is an **abstract shape representation** (for example, consonant-heavy, CV-like, repetitive, cluster-dense).

Stage 2: Family-level phonological modelling

For each language family, a coarse phonological profile is defined, including:

- Typical phoneme classes
- Common syllable templates
- Known phonotactic constraints
- Attested repair strategies in loanword adaptation

Profiles are intentionally broad and conservative.

Stage 3: Hypothetical phonologisation

The abstract symbol string is mapped into hypothetical phonological units under each family profile. No fixed sound values are assigned; instead, symbols are treated as placeholders that are constrained by family-level expectations.

Minimal repairs are applied to achieve legality under each profile.

Stage 4: Distortion assessment

Each hypothesis is evaluated according to:

- Number of constraint violations
- Severity of repairs
- Typicality of repairs for that family

The result is a **plausibility score**, representing relative phonological strain.

Stage 5: Cross-family comparison

Scores are compared across families to:

- Rank plausibility
- Identify incompatible families
- Highlight stable versus unstable hypotheses

Results are interpreted comparatively, not absolutely.

5. Illustrative example

Consider a hypothetical symbol string that, after structural abstraction, yields the following pattern:

C C C V C C

No specific sound values are assumed. Instead, the pattern is treated as an abstract structural shape and evaluated against different family-level phonological profiles.

Under cluster-tolerant profiles (for example, many Indo-European systems), such a pattern may be accommodated with relatively few repairs, since complex onsets and codas are attested and do not necessarily violate phonotactic constraints.

Under CV-preferential profiles (for example, many Austronesian systems), the same pattern would typically require systematic vowel insertion in order to satisfy syllable structure constraints. This results in higher phonological distortion relative to the original abstract pattern.

Under Semitic-like profiles, the pattern may be interpreted as consonantal material compatible with root-based morphological systems, where vowels are inferred or supplied by morphological templates rather than explicitly represented.

The framework does not assert that any of these interpretations is correct. Instead, it evaluates relative phonological compatibility by comparing the degree, type, and typicality of repairs required under each profile.

6. Strengths and limitations

Strengths

- Conservative and falsification-oriented
- Applicable to unknown or artificial systems
- Explicit about assumptions and scope
- Reusable across domains

Limitations

- Requires sufficient string length for stability
 - Operates at family, not language, resolution
 - Produces probabilistic, not deterministic, results
 - Cannot recover meaning
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7. Intended use cases

- Evaluation of competing phonological hypotheses
 - Assistance in undeciphered script research
 - Cross-linguistic phonological comparison
 - Exploratory analysis of artificial symbol systems
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8. Conclusion

This framework proposes a method for assessing phonological plausibility without overstepping into claims of decipherment or interpretation. By focusing on constraints, repairs, and comparative strain, it provides a structured way to narrow hypothesis spaces and to articulate why certain phonological assumptions are more plausible or less plausible than others.

The framework is intended as a complementary tool, not a replacement for existing linguistic or epigraphic methods.
